Score	IB Grade

Name: ..... **Test 4** **Mme Doux** *11th IB*

EXERCISE 1 **[14 marks]**

Let $f(x) = \dfrac{3x+4}{2x-5}$, for $x \neq \dfrac{5}{2}$.

(a) For the graph of $y = f(x)$, find the coordinates of:

(i) the x-intercept;

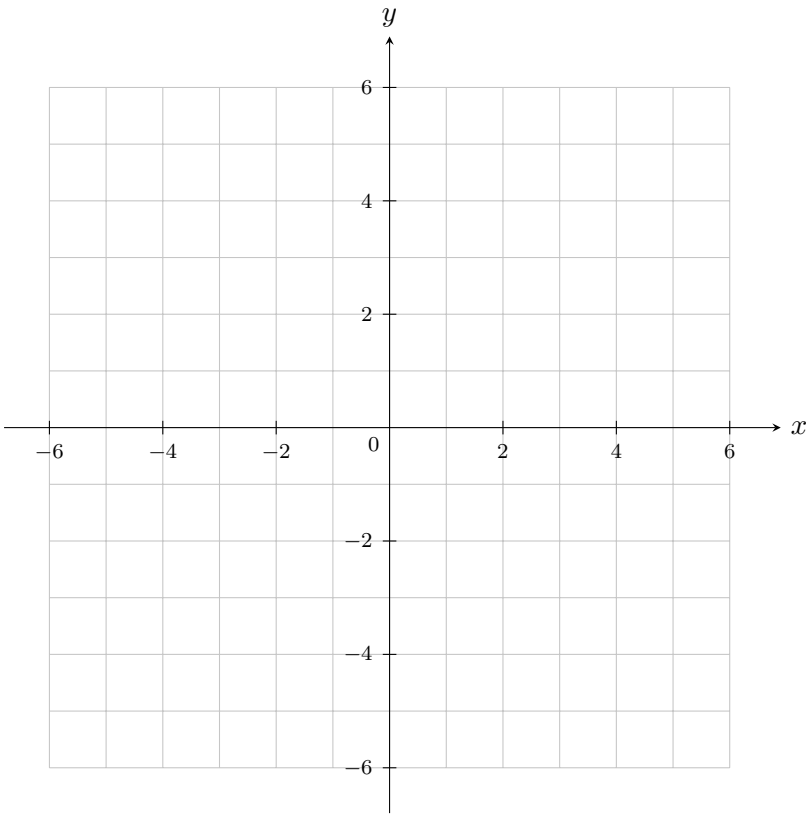
(ii) the y-intercept. **[3]**

(b) For the graph of $y = f(x)$, find the equation of:

(i) the horizontal asymptote;

(ii) the vertical asymptote. **[3]**

(c) Hence sketch the graph of $y = f(x)$. **[2]**



(d) Find $f^{-1}(x)$. **[3]**

The graphs of $f(x)$ and $f^{-1}(x)$ have two points of intersection.

(e) Find the coordinates of the two points of intersection. **Give your answers to two decimal places. [3]**

(BONUS) Hence find the equation of the perpendicular bisector to the two points of intersection. **[+2]**

EXERCISE 2 **[5 marks]**

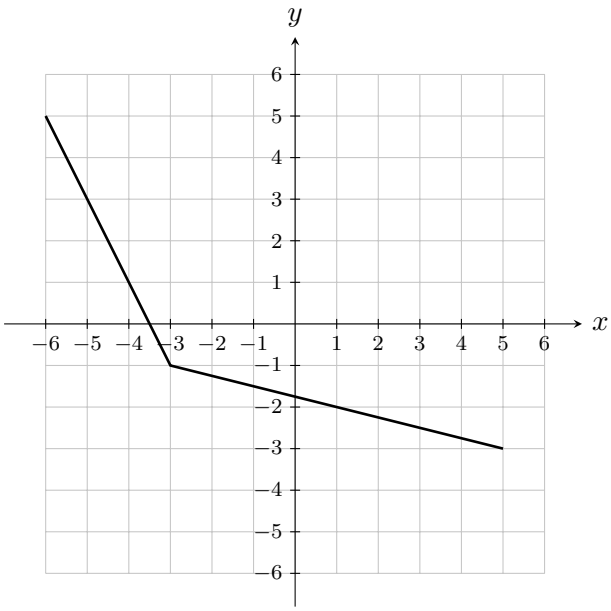
The functions f and g are defined such that $f(x) = \dfrac{x+3}{4}$ and $g(x) = 8x + 5$.

(a) Show that $(g \circ f)(x) = 2x + 11$. **[2]**

(b) Given that $(g \circ f)^{-1}(a) = 4$, find the value of a. **[3]**

EXERCISE 3 **[5 marks]**

The following diagram shows the graph of $y = f(x)$, for $-6 \leq x \leq 5$.



(a) Write down the value of $f(-3)$. **[1]**

(b) State the domain of f^{-1}, the inverse function of f. **[1]**

(c) Find the value of x that satisfies $f^{-1}(2x - 7) = -3$. **[3]**

END !